

Nucleic Acids - Factoid Sheet

Week 4 - Monday - Core Lecture 1 - current faculty deck + official Week 4 objectives

Core chain: duplex stability → genome organization → chromatin accessibility → RNA function. **Write every nucleic-acid sequence 5' to 3'.**

Denaturation / Annealing / Hybridization

Denaturation favored by Heat - extreme/high pH - low ionic strength - chemical denaturants. No phosphodiester bonds are broken.		T_m = 50% duplex melted. Higher GC → higher T_m. Melting causes a hyperchromic shift at 260 nm.	
Condition	Temperature	Salt	What pairs?
High stringency	High	Low	Only closely matched sequences
Low stringency	Lower	Higher	Related sequences with some mismatch

Prokaryotic DNA

Circular + negatively supercoiled + relatively naked. Gyrase = bacterial type II topoisomerase; transient DSB + ATP. Ciprofloxacin/novobiocin block gyrase resealing → DSB accumulation.

Eukaryotic Genome

Feature	Anchor
Nuclear DNA	Linear; introns/exons; telomeres with 3' overhang
Mitochondrial DNA	Circular
Highly repetitive	Centromeres/telomeres; tandem; usually inactive
Moderately repetitive	LINEs + SINEs; dispersed mobile-element related sequences
Minisatellite	VNTR polymorphism
Microsatellite	STR polymorphism; trinucleotide expansion disease

Sequence Basics

Backbone Phosphodiester bonds.	Pairing Antiparallel + Watson-Crick complementarity.	Read/write Always 5' to 3'.
--	--	---------------------------------------

Repeat Map

Type	Size/pattern	Use / risk
Highly repetitive	5-500 bp tandem	Centromere/telomere; silent
LINE	~5-6 kb dispersed	Retrotransposon-related
SINE	~100-200 bp dispersed	Retrotransposon-related
Minisatellite	Usually 9-30 bp x 10-60	VNTR polymorphism
Microsatellite	2-6 bp x tandem	STR; repeat-expansion disease

Chromatin

Nucleosome 2 each H2A/H2B/H3/H4 + ~150 bp wrapped DNA. H1 binds linker DNA.	State Euchromatin = open/active. Heterochromatin = condensed/silent.
---	---

Epigenetics

Change	Enzyme	Typical result
CpG methylation	DNMT	Compaction / silencing
Histone acetylation	HAT	Open / active
Deacetylation	HDAC	Closed / repressed
H3K4me3	-	Activation anchor
H3K27me3	-	Repression anchor

Nucleic Acids - Fast Recall

Epigenetic Drug Anchors

Decitabine DNMT inhibitor → less DNA methylation → tendency toward gene re-expression.	Valproic acid HDAC inhibitor in lecture → more histone acetylation → tendency toward open chromatin.
--	--

RNA Classes

RNA	Numbers / structure	Function
tRNA	75-85 nt; modified bases; 3' CCA; cloverleaf → L shape	Adaptor; carries amino acid + anticodon
rRNA	Prok: 23S/16S/5S; Euk: 28S/18S/5.8S/5S	Ribosome; 23S/28S peptidyl transferase
mRNA	Euk 5' m7G cap + 3' poly(A); bacterial often polycistronic	Protein-coding message
hnRNA	Precursor RNA	mRNA precursor
snRNA	U-rich, nuclear	Pre-mRNA maturation/splicing
scRNA	Cytosolic + proteins	Signal-peptide recognition pathway
lncRNA	>200 nt	Chromatin/gene regulation; Xist silences X
miRNA	~19-25 nt	Post-transcriptional silencing

RNA Abundance / Prok-Euk Contrast

Approx abundance mRNA 2-5% - tRNA 10-15% - rRNA 75-80% of cellular RNA.	mRNA contrast Euk: cap + poly(A), usually monocistronic. Prok: lacks those classic euk processing features, commonly polycistronic.
---	---

Must-Not-Miss Traps

High stringency = high temp + low salt.	Hybridization can bind related sequences; poor default tool for a single-base mutation.
Histone methylation is context specific; do not use the simple "methyl closes" shortcut for histone tails.	Lys acetylation neutralizes positive charge and loosens DNA binding.

Drug / Disease Anchors

Ciprofloxacin + novobiocin → gyrase.	Decitabine → DNMT inhibitor.
Valproic acid → HDAC inhibitor in deck.	Trinucleotide repeat expansion → microsatellite instability mechanism/disease anchor.

30-Second Review

Tm rises with GC. Gyrase supercoils bacterial DNA. Telomeres/repeats characterize eukaryotic chromosomes. Euchromatin open; heterochromatin closed. DNMT closes; HAT opens; HDAC closes. tRNA adapts; rRNA catalyzes; mRNA encodes; Xist silences; miRNA represses.

Assigned ScholarRx

DNA Structure: <https://exchange.scholarrx.com/brick/dna-structure>